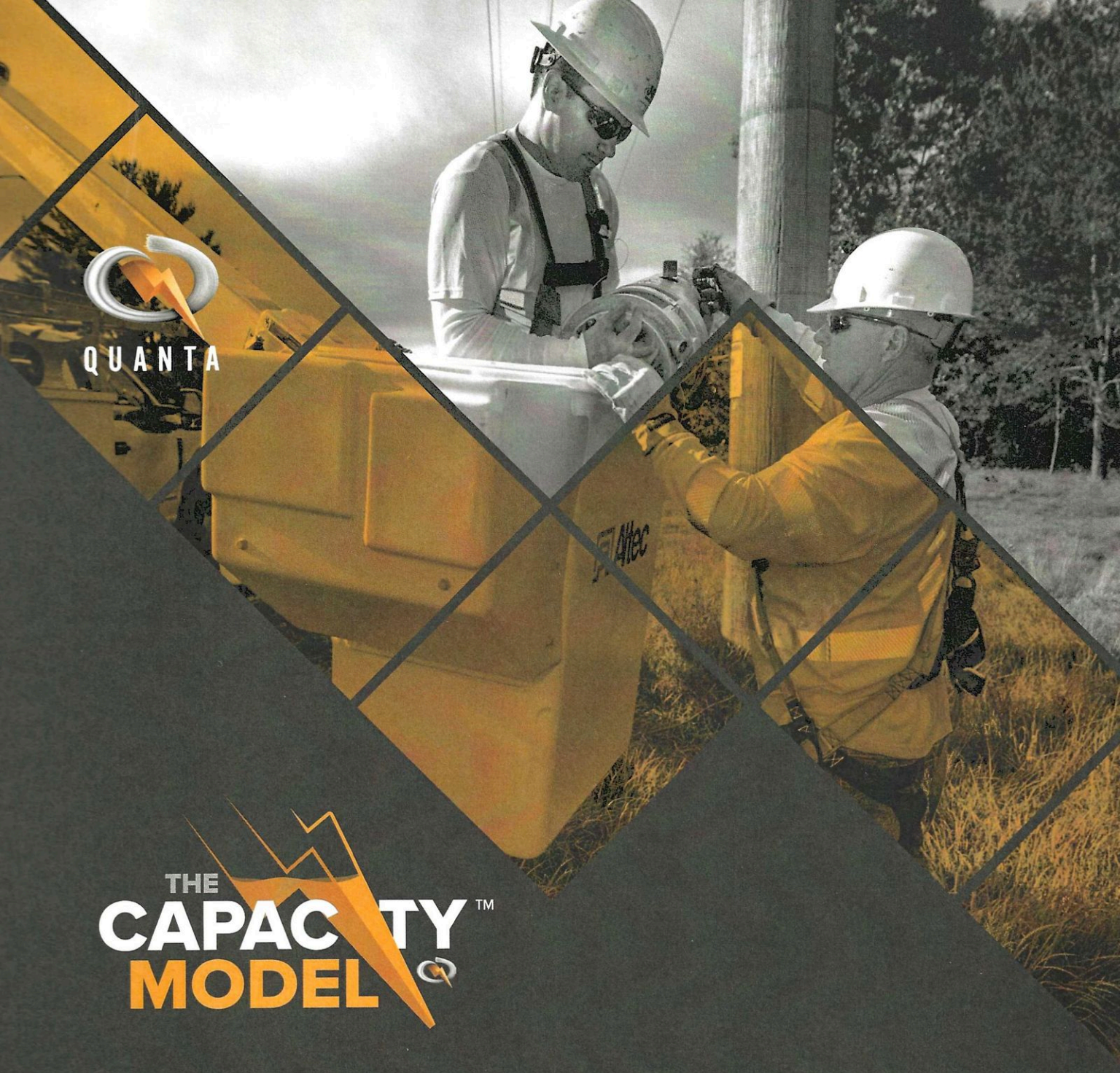

QUANTA

THE CAPACITY MODEL™

STKY AND THE ENERGY WHEEL

FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

TRAINEE MANUAL

V 1.0

NLC

THANK YOU

This course is the product of long months of research, discussion, planning, real-world application, more discussion, and site visits with workers and leaders in the utility industry worldwide. The key principles of The Capacity Model® are based on the research of international experts in workplace safety, human performance, risk management, and other fields. We are grateful for their outstanding contributions to the industry and for their cooperation with our team.

Our goal is to improve how all workers and executives approach safety in the workplace—and to bring more people home safely to their families every night.

We'd like to thank the dedicated individuals and organizations who have devoted so much of their time and effort toward making this training a reality.

Quanta Services Corporate SH&E Department

Quanta Operating Companies

Individual Experts in Industrial Safety

THE CAPACITY MODEL™

CONTENTS

SYLLABUS **4**

AGENDA **5**

CH. 1 STKY: STUFF THAT KILLS YOU **7**

1.1 STKY: Stuff That Kills You 8

1.2 Controls 9

1.3 STKYa 11

1.4 STKYp 12

1.5 STKY Success 13

ACTIVITY: STKY Event Classification Tool 14

CH. 2 THE QUANTA ENERGY WHEEL **17**

2.1 Identifying Hazards 18

2.1.1 Blind Spots 18

2.1.2 The Energy Wheel: An Effective Tool 22

2.2 Using the Energy Wheel 24

2.2.1 The Energy Wheel and STKY Discussions 25

2.2.2 The Energy Wheel and Job Briefings 27

2.2.3 Asking Capacity Questions 29

ACTIVITY: Job Briefing 40

GLOSSARY **51**

SOURCES **53**

APPENDIX **55**

STKY and the Energy Wheel for the Power and Telecommunications Industry

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TCM 3004

STKY AND THE ENERGY WHEEL FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

SYLLABUS

COURSE DESCRIPTION

Trainees learn how to identify events that result from the release of energy and to identify and categorize workplace hazards.

The class is taken in conjunction with the Quanta Human Performance Principles module. Complete both modules within three months to gain maximum learning and retention benefits.

Intended audience: Operations management, safety professionals, supervisors who direct the work of others, and those responsible for leading job briefings.

LEARNING OUTCOMES

1.a Define a STKY control.
1.b Define a STKY event, STKY potential, STKY actual, and a STKY success.
1.c Apply the STKY Event Classification Tool to practical scenarios.
2.a Use the Energy Wheel to identify hazards associated with a task.
2.b Identify preventive and mitigative controls to prevent and reduce the consequences of unwanted events.

SUGGESTED PREREQUISITES

- Understanding The Capacity Model®

COURSE MATERIALS

- Materials to be issued at the beginning of the course.

COURSE OUTLINE

- Housekeeping
- STKY
- The Energy Wheel
- Exam

INSTRUCTIONAL METHODS

- Lecture
- Learning activities
- Knowledge check

METHODS OF MEASUREMENT

KNOWLEDGE
Written exam:
A = 100%–90%
B = 89%–80%
C = 79%–70%
D = 69%–60%
F = 59% and below

If a student earns less than a 70% on the exam, one retake is permitted. If they do not earn at least a 70% on the exam retake, they must retake the course.

SKILL
Completion of all activities as directed by the instructor.

BEHAVIOR
Active participation and engagement in class discussions and activities.

STUDENT-TO-INSTRUCTOR RATIO
Maximum of 24:1.

CLOCK HOURS
5 Hours

POLICIES AND EXPECTATIONS

Building a positive learning environment—Be sure to share, support your classmates, and contribute to the positive learning environment during your course. You bring incredible value to the group and to the course. Please refrain from any behavior that you know doesn't work in a classroom environment.

TCM 3004

STKY AND THE ENERGY WHEEL FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

AGENDA

SCHEDULE

HOUSEKEEPING	*15 minutes*
STKY	*1 hour, 30 minutes*
ENERGY WHEEL	*2 hours, 45 minutes*
EXAM	*30 minutes*

Breaks are scheduled at the discretion of the instructor.

The lunch schedule is determined by the class start time and the option to complete the Quanta Human Performance Principles module in the same day.

STKY: STUFF THAT KILLS YOU

LEARNING OUTCOMES

After completing this chapter, you should be able to:

1.a Define a STKY control.

1.b Define a STKY event, STKY potential, STKY actual, and a STKY success.

1.c Apply the STKY Event Classification Tool to practical scenarios.

1.1 STKY: STUFF THAT KILLS YOU

STKY (pronounced "sticky") stands for *stuff that kills you* (you may hear it referred to as "s**t that kills you"). STKY includes a source of energy that, when released and not controlled, can be:

- Life-threatening (requires medical intervention to survive).
- Life-altering (results in loss of a body part or function).
- Life-ending (results in the death of the worker).

You may be familiar with SIF (serious injuries and fatalities). STKY seeks to replace SIF's more academic definition with a phrase that is provocative, attention-grabbing, and easy to remember.

STKY is a foundational concept that drives The Capacity Model®. It's what we want to focus on because, after a steady decline for many years, the number of life-ending workplace events leveled out before beginning to increase again in recent years. A focus on STKY is key to understanding all other elements of The Capacity Model.

A **STKY event** is an event where preventive controls failed, resulting in the unexpected or uncontrolled release of energy that could have reasonably resulted in a life-threatening, life-altering, or life-ending event.

For reference, the industry would classify a STKY event as a potential serious injury or fatality (SIFp). A STKY event results in one of the following three scenarios:

- **STKYa:** A stands for "actual." A work incident (STKY event) that resulted in an injury that was life-threatening, life-altering, or life-ending.
- **STKYp:** P stands for "potential." A work incident (STKY event) in which no one was seriously injured, but mitigative controls were either not in place, not functioning, or not effective. The only thing protecting the crew from STKYa was luck.
- **STKY success**: A situation where mitigative controls were in place, functioning, and effective. These controls successfully prevented a STKY event from becoming a STKYa.

1.2 CONTROLS

Before looking more closely at the types of STKY events, we should define the term *control*. In the industry, a **control** refers to acts (things a person does), objects (a device that works without acts), or systems (a combination of acts and objects) intended to prevent or mitigate an unwanted event.

Preventive controls are acts, objects, or systems designed to prevent an unwanted event. When evaluating a preventive control, ask the question, "How does the control prevent the unwanted event?"

Examples include:

- Maintaining a minimum approach distance from an energized part.
- De-energizing a line.
- Completing the pre-travel checklist.
- Using wheel chocks.

Mitigative controls are acts, objects, or systems designed to reduce the consequences of an unwanted event. These controls are critical because systems break, equipment fails, and people make mistakes. When evaluating a mitigative control, ask the question, "How does the control minimize the consequences after the unwanted event occurs?"

Examples include:

- Wearing PPE like hard hats, safety glasses, and insulated gloves.
- Using a fall restraint system.
- Establishing an exclusion zone around the fall zone of an elevated work area.
- Posting a qualified observer at the entrance to a confined space.

A **STKY control** is a control that prevents or mitigates a STKY event (life-threatening, life-altering, or life-ending event). Like preventive and mitigative controls, it is an act, object, or system. STKY controls create the capacity to fail safely and ensure workers are safe to begin work. STKY controls must:

1. Specifically target the energy source.
2. Effectively prevent and mitigate exposure to the energy source when installed, verified, and used properly (so a STKYa cannot reasonably occur).
3. Work even if there is human error during the work period (unrelated to the installation of the control).

Review the following example:

Task: Trenching

Energy source: Motion (collapse)

Preventive and mitigative controls targeting the energy source:

- Assess the area and identify soil conditions using a qualified and competent person.
- Apply support systems such as trench shields, shoring, bracing, or underpinning.
- Establish adequate egress and escape routes.
- Inspect trenches at the start of each shift, after a rainstorm or water event, and after any event that could potentially change the conditions in the trench.

Fig. 1.1. Put STKY controls in place to prevent and mitigate STKY events.

1.3 STKYa (ACTUAL)

STKYa (actual) is a STKY event that resulted in an injury that was life-threatening, life-altering, or life-ending:

- *Life-threatening*—A state where death is imminent or probable if not immediately addressed by trained medical personnel, often with the aid of life-sustaining support. Examples include profuse bleeding, damaged or blocked airway, untreated shock, or unnatural arrhythmia (irregular heartbeat due to unnatural causes like a shock or ingestion of a toxic substance).
- *Life-altering*—A state of permanent or long-term impairment or loss of the use of an internal organ, bodily function, or body part. Example injuries include amputations, loss of vision or eye, compound fractures, spinal injuries or paralysis, severe burns, and disfigurement.
- *Life-ending*—An injury that results in the death of an employee.

PHOTO BY THOMAS KAROL FOR UNITED STATES AIR FORCE

Fig. 1.2. STKY actuals are life-threatening, life-altering, and life-ending events.

1.4 STKYp (POTENTIAL)

STKYp (potential) is a STKY event in which no one was seriously injured, but mitigative controls were either not in place, not functioning, or not effective. The only thing protecting the crew from STKYa was luck. The crew did not have capacity for failure.

Here are two examples of STKYp:

- The rigging on an elevated load fails and the load drops to the ground. No exclusion zone was set up. Luckily, none of the workers nearby were injured.
- A crew is using a backhoe to dig a 5-ft-deep trench around some buried utility lines. Before a trenching box is put in place, a worker enters the trench to check the condition of the utilities. After a few minutes, he notices a growing bulge in the trench wall as some soil breaks off onto the trench floor. The worker scrambles to safety shortly before the trench wall collapses.

In both situations, the only thing preventing a serious injury was luck.

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Fig. 1.3. Implement mitigative controls , like an exclusion zone, to avoid STKYa and STKYp events.

1.5 STKY SUCCESS

A **STKY success** refers to a STKY event where mitigative controls were in place, functioning, and effective. These controls prevented a STKY event from becoming a STKYa. The mitigative controls protected employees from the unwanted release of energy. The crew had capacity for failure.

To be considered a STKY success, the two STKYp scenarios would instead play out like this:

- The rigging on an elevated load fails and the load drops to the ground. The workers set up an exclusion zone before the lift so the crew was not in the lift area when the load dropped.
- A crew is using a backhoe to dig a 5-ft-deep trench around some buried utility lines. A trenching box is properly installed in the trench before a worker enters to check the condition of the utilities. After a few minutes, the soil wall beyond the trenching box unexpectedly collapses but the trenching box protects the worker from a cave-in.

A GOOD CATCH

A good catch is sometimes mislabeled as a STKY potential or a STKY success. A **good catch** is recognizing a situation or circumstance which could have led to a STKY event, but thanks to timely corrective action or intervention, there was no unexpected or uncontrolled release of energy. Some examples include:

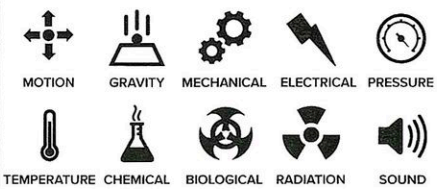
- *Recognizing a cracked weld on the boom of an aerial lift before placing it in service.* An unwanted release of energy did not occur because the cracked weld was caught and the lift was removed from service.
- *Discovering mechanical or structural failures of equipment during a pre-trip inspection.* An unwanted release of energy did not occur because the failure was discovered before placing the equipment in service.
- *An install crew up in a tower notices a loose nut lying near an unprotected edge. A crewmember picks up the loose nut.* An unwanted release of energy did not occur because the nut was secured before it was accidentally kicked off the edge.

STKY EVENT

CLASSIFICATION TOOL

1.
STKY EVENT

Preventive controls failed resulting in the unexpected or uncontrolled release of energy that could have reasonably resulted in a life-threatening, life-altering, or life-ending event.



2.
STKY ACTUAL

Was there an event that resulted in an injury that was life-threatening, life-altering, or life-ending?

3.
CONTROLS

Were there adequate mitigative controls in place that targeted the energy, that were installed and used properly, and that functioned correctly despite human error?

THERE WAS AN EVENT!

STKY EVENT

CLASSIFICATION TOOL SCENARIOS

USE THESE SCENARIOS TO PRACTICE CLASSIFYING STKY EVENTS. DETERMINE IF EACH SCENARIO IS A STKYa, STKYp, STKY SUCCESS, OR NOT STKY. CIRCLE THE APPROPRIATE CLASSIFICATION.

1. A forklift operator was loading empty reels into the back of a truck. The truck driver, not realizing his vehicle was still being loaded, tried to pull away. The loading dock restraint system stopped the truck from moving. The forklift operator was able to move safely out of the truck bed.

   STKYa **STKYp** **STKY success** **Not STKY**

2. An employee was working on a ledge approximately 3 ft high when he fell, resulting in a bruised hip and a cut that required four stitches.

   STKYa **STKYp** **STKY success** **Not STKY**

3. A crew was changing out a utility pole. As the digger derrick operator was moving the pole into place, a crewmember narrowly missed being struck by the pole. There was not a marked exclusion zone.

   STKYa **STKYp** **STKY success** **Not STKY**

4. An employee was working on a 10-ft ladder, which another worker stabilized from below. The worker on the ladder accidentally dropped a tool that hit the lower worker in the face. Injuries included a fractured orbital bone and permanent vision loss in her right eye.

   STKYa **STKYp** **STKY success** **Not STKY**

5. A crew responded to an emergency call to repair a line. Before ascending the pole, one employee was inspecting her work positioning equipment when she noticed a cut and fraying on her positioning strap. She removed the strap from service.

   STKYa **STKYp** **STKY success** **Not STKY**

THE QUANTA ENERGY WHEEL

LEARNING OUTCOMES

After completing this chapter, you should be able to:

2.a Use the Energy Wheel to identify hazards associated with a task.

2.b Identify preventive and mitigative controls to prevent and reduce the consequences of unwanted events.

2.1 IDENTIFYING HAZARDS

2.1.1 BLIND SPOTS

Every job site poses multiple threats to worker safety. Some hazards are obvious to workers, like falling from an elevated position or being hit by falling equipment. Other hazards are sudden and unexpected, such as a rigging failure or a passing vehicle losing control. Still other hazards are dangerous because they occur slowly or invisibly, such as dehydration or exposure to toxic fumes.

For someone with a job to do, it's a challenge to account for every possible hazard and then keep track of those hazards during the workday. Research has shown that workers in the construction trade can consistently identify only about 45% of workplace hazards at any given time.[1,2] With this low level of awareness common throughout the industry, it's not a matter of *if* a crew will face a serious incident, but *when* it will happen.

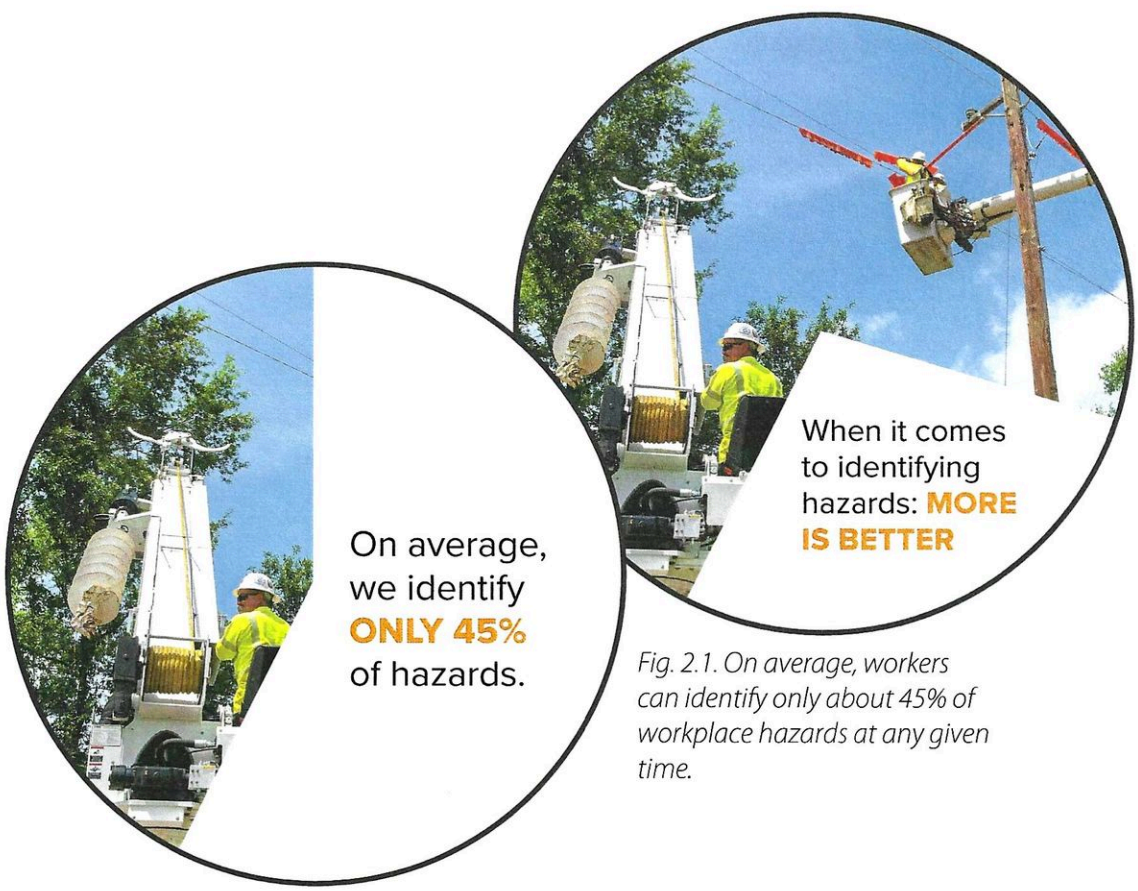


Fig. 2.1. On average, workers can identify only about 45% of workplace hazards at any given time.

SENSORY MANAGEMENT

A worker's inability to notice hazards is rarely caused by ignorance or lack of effort. This mental "blind spot" is a natural side effect of the brain filtering the massive amount of information that enters through the five senses.[3] As an example, imagine standing in a sunny field. Your body is sensing countless things—the heat of the sun on your face, the sound of nearby songbirds, the sight of individual blades of grass moving with the wind, the taste of the gum in your mouth, the smell of a distant barbeque. Just those five inputs might be too much for your brain to think about all at once, so it must prioritize which ones to focus on.

Now compare this peaceful setting to a busy job site, with major hazards demanding your attention from every direction. Sometimes it's amazing workers notice as much as they do. The only way for the brain to boost a worker's focus on a certain task or hazard is to suppress other sensory inputs.[4]

Fig. 2.2. The human brain filters out massive amounts of information in order to function.

SIMPLE VERSUS COMPLEX THINKING

Another factor for how well a worker notices a hazard is the type of energy involved. This matters because our brains react to different situations in different ways. For example, the medulla is the instinctual part of the brain. It is responsible for involuntary activities, such as managing breathing, swallowing, and heart rate.[5] The medulla also makes our body react involuntarily to things like a falling motion (when you instantly grab something to steady yourself) or sudden movement (closing your eyes and ducking when something flies near your head). Because hazards like falling or motion trigger our danger instincts, we notice them more easily. [1]

But what about less obvious hazards, like chemicals, radiation, or electricity? These may not appear dangerous unless a worker thinks about them specifically. Recognizing these as hazards requires more thought—and a different part of the brain—than instinctual risks like a utility pole or pipeline segment swinging near your head.[1]

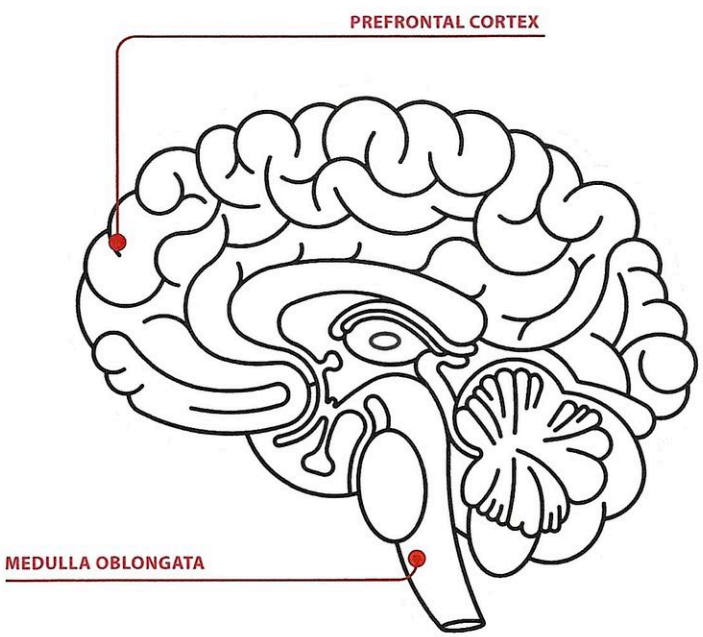


Fig. 2.3. Different parts of the brain process different types of information.

The part of the brain called the prefrontal cortex controls more complex behaviors like focusing attention, organizing, and short-term memory.[6] This is the section of the brain that must activate for a worker to recognize, for example, that a metal cable on the ground may be energized. The worker must notice the cable, see that it's metal, wonder if it's attached to an energy source, and then track down whether it is actually an electrical risk. This sort of multistep, slightly complex investigation is the sort of thing a worker's brain will not do naturally. The human brain tends to see the instinctual hazards first and may skip over more complex hazards that are equally dangerous.

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Fig. 2.4. Your brain instinctually sees some situations as more hazardous than others.

The challenge is that workers must be able to notice or recognize all types of hazards if they are to protect themselves. However, if workers notice less than half of the energies on the job site, they remain highly exposed to life-threatening, life-altering, or life-ending hazards. To overcome the brain's natural function, workers need some help.

2.1.2 THE ENERGY WHEEL: AN EFFECTIVE TOOL

The number of hazards that a worker will notice jumps from around 45% to more than 70% when the worker uses an energy-based approach to look for hazards.[7,1] In other words, if a worker systematically examines a job site based on the types of energy, that worker will notice more hazards. That's where the Quanta Energy Wheel comes in.

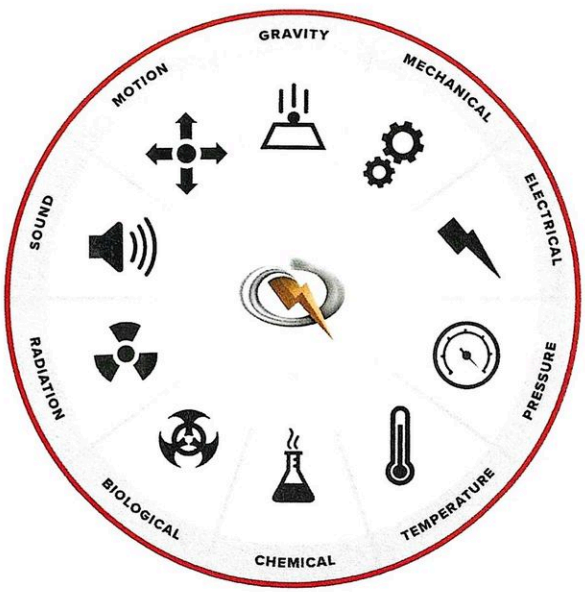

Fig. 2.5. The Quanta Energy Wheel.

The Energy Wheel in figure 2.5 is a job aid used to identify energies before and during work. It organizes hazards into 10 categories.

The most easily recognized energy hazards are at the top of the Energy Wheel, beginning with motion. They are strategically placed in order of most commonly identified energies to least commonly identified as you move clockwise around the wheel, ending with sound.

Table 2.1 lists the energy categories and includes two definitions. One is a more scientific definition that workers and safety personnel should understand; the other is a "shoutable" definition that is easy to remember—and easy to shout at someone in a work environment.

ENERGY	SHOUTABLE	DEFINITION
MOTION	Stuff that moves.	A change in position of objects or substances.
GRAVITY	Stuff that falls.	Force caused by the gravitational attraction of all objects to the earth.
MECHANICAL	Stuff that twists, turns, or pounds.	Rotation, vibration, or motion within a moveable part of a fixed or stationary piece of equipment.
ELECTRICAL	Stuff that shocks.	The presence of electrical energy (current).
PRESSURE	Stuff that's stressed.	The force that a gas, liquid, or solid exerts on another object.
TEMPERATURE	Stuff that's hot or cold.	Differences in thermal energy of objects or the environment with the human body.
CHEMICAL	Stuff that's toxic.	Substances that inherently or through reaction have the potential to create a physical or health hazard.
BIOLOGICAL	Stuff that's living.	Living organisms and the natural environment that can present a hazard.
RADIATION	Stuff that's radioactive.	Elements that emit ions or atomic particles.
SOUND	Stuff that's loud.	Hazardous sound levels for hearing and operations.

Table 2.1. Components of the Energy Wheel.

2.2 USING THE ENERGY WHEEL

The Energy Wheel focuses a crew's attention on one energy at a time to make hazards easier to notice. As an example of how this would work, look at Figure 2.6. There is a lot going on in this photo. Instead of just asking, "What do you see?" you can use more specific questions like, "How many yellow cabs do you see?" or "How many traffic lights do you see?" Noticing details in a busy setting is much easier when your attention is focused on one item at a time. The Energy Wheel helps you and the crew identify and visualize hazards one energy at a time.

PHOTO BY AHUNDT ON PIXABAY

Fig. 2.6. How many FedEx trucks do you see? The Energy Wheel targets attention on a job site just like specific questions target your attention on this image.

The Energy Wheel should be used on all jobs—from large jobs with multiple tasks in a constantly changing environment to small, *seemingly* routine jobs. Remember that the Energy Wheel is a tool to systematically identify hazards. It is not meant to categorize the hazards already identified. While you and the crew will focus on each energy to identify hazards, recognize that every energy will not be present for every job on every job site.

While the Energy Wheel can be used to identify all hazards, crews want to focus on STKY: life-threatening, life-altering, or life-ending events. The hazards that cause a sprained ankle are rarely the same as the ones that cause a fatality. Workers can recover from a sprained ankle—they cannot recover from a life-ending event.

2.2.1 THE ENERGY WHEEL AND STKY DISCUSSIONS

STKY discussions are conversations meant to generate an understanding of how to identify STKY hazards using the Energy Wheel. These should be part of normal job briefings at the start of the day, the start of a job, and anytime conditions change that may impact worker safety.

Recommendations for a STKY discussion:

- Hold informal conversations.
- Create a safe space where workers feel comfortable sharing their thoughts and making mistakes while they learn to identify hazards.
- Rather than calling out hazards to crewmembers, encourage the crew to participate in hazard identification using the Energy Wheel.
- Focus only on identifying hazards. Avoid compliance-based training during this time.
- Use the following questions to get your STKY discussion started:

    1. What is STKY on this job site? (The Energy Wheel is a tool to help identify hazards.)
    2. What will prevent a STKY event?
    3. When a STKY event happens, what will protect you and the crew?
    4. Is that protection enough, or do you need more? What else can you do to protect against STKY?

If you find your crew in the field with inadequate controls—or with insufficient layers of controls—stop work until adequate controls can be put in place. "Stop If Unsure" means you and your crew have an obligation to stop work and escalate to leadership when there is a gap in controls. Although workers always have an obligation to stop work, do not depend solely on a person's judgment to recognize when to stop work. "Stop If Unsure" will not guarantee worker safety. Instead, plan for and execute work with the adequate STKY controls to keep workers safe and don't start until those controls are verified to be in place and functioning.

Training crews to identify STKY easily and correctly takes time—it requires workers to change how they think of their job site. No matter how good the first STKY training is, regular STKY discussions and consistent leadership are required to create lasting change.

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Fig. 2.7. Training employees to recognize STKY takes time and consistent practice.

2.2.2 THE ENERGY WHEEL AND JOB BRIEFINGS

Every job begins with a **job briefing** (sometimes called a *tailboard* or *toolbox talk*). Job briefings are meetings led by the crew leader and involve the entire crew. They are held at the beginning of the job and as needed during the day. Job briefings ensure every crewmember understands:

- The scope of the job.
- The tasks assigned to each crewmember.
- The hazards involved and what will be done about them. Use the Energy Wheel as a tool to identify the hazards.
- What equipment is being used, and who is using it.
- What protective equipment must be used.
- Emergency procedures (such as who will call 911 in case of an emergency).

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Fig. 2.8. Job briefings must involve the entire crew.

Every member of the crew must be involved in the job briefing. If someone shows up late or a visitor arrives, they should hear all the information covered in the job briefing. If someone leaves the job site, or if conditions change, another job briefing may need to be held to adjust plans and assignments. Follow-up briefings may be held (such as after a lunch break) to remind the crew of upcoming tasks and hazards.

Fig. 2.9. Crew leaders help workers respond to challenges in meaningful ways.

Job briefings can become a meaningless formality if they are treated that way. If the crew uses that time to organize themselves, ask questions, improve safety, and look for ways to work more efficiently, the job briefing can be meaningful and productive. Use the Energy Wheel during all job briefings. The wheel's structured approach will help the person conducting the briefing and each employee notice hazards they may otherwise miss.

If a near miss occurs, that's an excellent moment to hold a job briefing. Pause the work to figure out what just happened and how. People involved may be feeling angry or defensive—this can degrade your briefing into a shouting match of accusation and insults. To avoid this result, the crew leader asks questions that guide the crew away from blaming and toward learning.

The goal is to move beyond pointless, knee-jerk reactions to mistakes. Instead, respond to the situation in a meaningful way that focuses on controls. If the crew walks away from the briefing knowing something will be improved to make them safer (such as a system or tool), it promotes confidence and a willingness to move forward with the job.

2.2.3 ASKING CAPACITY QUESTIONS

Whether you are holding a STKY discussion or working through the hazard identification portion of the job briefing, you will guide your crew to the Energy Wheel to identify hazards on the job site. For every hazard that you and your crew notice, you must then identify unwanted events that can stem from that hazard. An **unwanted event** is an event with a potential for harm from exposure to a hazard that was not adequately controlled—it is the bad stuff that can happen because of a hazard. For example, gravity can have multiple unwanted events at a job site. There may be a potential for falling tools, dropped loads, and workers falling from elevated surfaces.

Once those unwanted events are identified, the crew must then ask capacity questions that answer: *What are we going to do about it?* During STKY discussions and job briefings, **capacity questions** help crews develop and implement preventive and mitigative controls specific to an unwanted event. Examples include:

- What alert systems are in place if any of the energies are released?
- If one energy is released, will that trigger a chain reaction of other energies?
- How do we protect against those additional energies?
- Is this control enough, or do we need more?

The actual questions will vary based on location, season, industry, time of day, and many other factors. Help crews learn how to tailor their questions based on current conditions.

The following pages dedicate a section to each energy on the Energy Wheel. Each section provides examples of unwanted events, capacity (controls) for the unwanted events, and sample capacity questions.

STUFF THAT MOVES

MOTION

Motion hazards are presented by moving objects, typically vehicles or heavy machinery. They also represent hazards related to trench cave-ins, loose material collapse or other movement, or any moving object that can generate enough force to create a hazard (such as shrapnel or debris from broken equipment). Capacity for motion hazards may include exclusion zones, PPE, or barriers.

Sample capacity questions for motion:

- Where are the suspended loads?
- Which equipment operators will use spotters?
- Where might our vehicles encounter pedestrians?
- Which vehicles have no backup alarm?
- When raising and lowering booms, who's watching the clearances?
- How is equipment swing radius marked or barricaded?
- Which trenches need to be shored?
- Which vehicles need the wheels chocked?
- Where will loose materials (such as poles, pipes, transformers) be stored and secured?
- How do we help workers observe buffer zones between themselves and moving equipment?

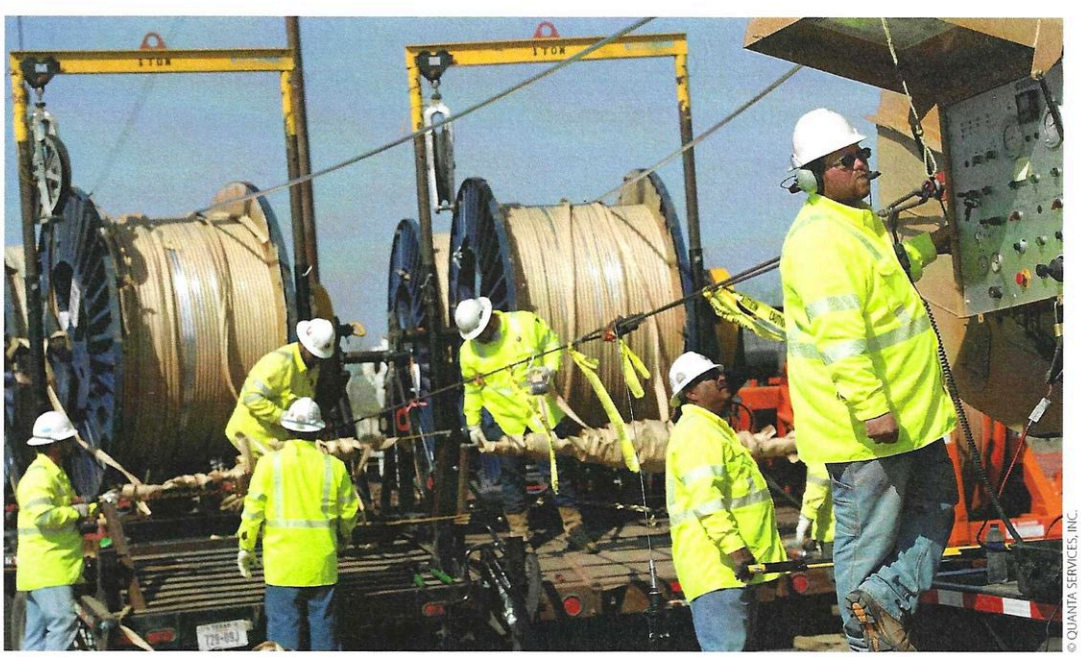

Fig. 2.10. Motion refers to any moving object that may impact a worker.

GRAVITY

The most common examples of gravity are suspended loads or objects, or personnel working from heights. A dropped load or a fall from height could quickly become a STKY event. Capacity for each might include an exclusion zone for suspended objects, and a safety harness for elevated workers.

STUFF THAT FALLS

Sample capacity questions about gravity:

- Where are the suspended loads?
- What safety gear is needed for the workers in elevated positions?
- Which vehicles are carrying loose loads?
- How will we secure the equipment stowed on scaffolding or upper levels?
- Who's establishing the exclusion zones?
- Where is the line of fire?

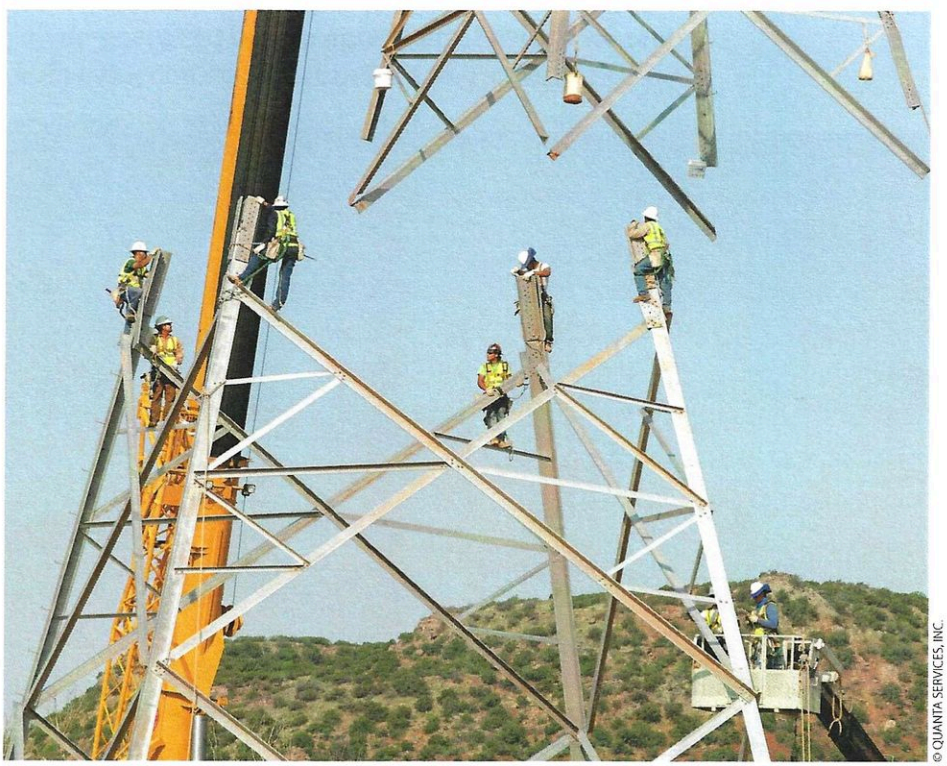

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Fig. 2.11. Gravity includes a falling worker and things that can fall onto a worker.

STUFF THAT TWISTS, TURNS, OR POUNDS

MECHANICAL

Mechanical hazards are typically associated with moving machinery like jackhammers, drills, or pinch points on heavy machines. Capacity for these varies depending on the type of mechanical energy, but can include exclusion zones, PPE like steel-toe boots, and kill-switches on the machines themselves.

Sample capacity questions about mechanical:

- Who's checking that all mechanisms are covered by protective shields?
- How will we ensure kill-switches are clearly marked and reachable?
- Who's establishing the exclusion zones?
- Who's supervising the outriggers during movement?
- How will workers know to keep their distance from operating augers, drills, and gears?
- Which workers are wearing steel-toe boots?

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Fig. 2.12. Mechanical hazards often involve a worker getting caught in operating machinery.

ELECTRICAL

The most common electrical hazards are energized lines. Frayed wires on equipment, overloaded circuits, and loose or improper connections are also possible electrical hazards. Electrocutions, arc flashes, and electric fires (even explosions) may result from working with or around energized electrical systems. Capacity for these hazards may include non-conductive PPE and rubber matting.

STUFF THAT SHOCKS

Sample capacity questions for electrical:

- How do we protect workers from second point of contact?
- How do we ensure enough protective cover-up in the work area?
- Where should we place barriers to protect against both step and touch potential?
- How do we know if the PPE is sufficient for the voltage on the job site?
- Who's checking that grounds are installed correctly?
- Who's verifying that dielectric equipment is current on inspections?
- If working on de-energized lines or equipment, how will we protect ourselves from induction, backfeed, or lightning strikes?
- In a plant environment, how do we check that circuits are de-energized and properly locked, tagged out, and verified?

Fig. 2.13. Electrical hazards exist whenever a crew works on a line that is live or one that is de-energized and accidentally becomes energized.

STUFF THAT'S STRESSED

PRESSURE

Pressurized systems such as gas cylinders, pipelines, and heavy equipment tires are common examples of pressure hazards. Any of these could be deadly if improperly handled or punctured. Capacity for these hazards typically includes exclusion zones and PPE, such as eye and ear protection.

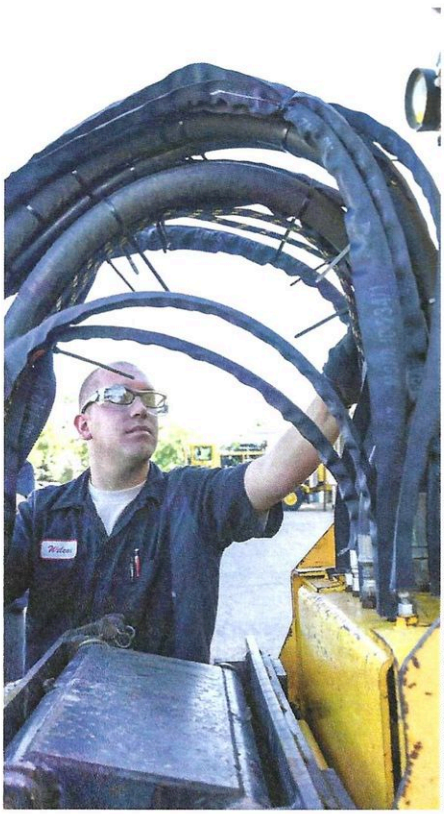

PHOTO BY CYNTHIA GRIGGS FOR UNITED STATES AIR FORCE

Fig. 2.14. Check for worn hoses that might release pressurized fluids if they fail.

Sample capacity questions for pressure:

- Which hoses need to be checked for wear, cuts, corrosion, or other damage?
- What is the condition of vehicle tires (such as wear or puncture risk)?
- How do we ensure that nozzles and hoses are attached securely to pressurized containers?
- Where will we store pressurized containers to keep them away from work areas?
- Who's establishing the exclusion zones?
- How do we control the energy released if a high-pressure hose or connector fails?

TEMPERATURE

Temperature-related hazards typically refer to ambient or seasonal temperature conditions (heat in summer, cold in winter) but can also refer to working with hot or cold equipment such as welding equipment or liquid nitrogen. Capacity for these hazards can include season-specific work attire, heaters and air conditioners, insulated coverings, and heat shields.

STUFF THAT'S HOT OR COLD

Sample capacity questions for temperature:

- What sort of clothing do workers need for this weather?
- How do we keep workers hydrated?
- How do we watch each other for weather-related ailments (such as hypothermia and overheating)?
- What PPE do workers need as insulation against hot or cold equipment?
- Who's maintaining exclusion zones around welding operations?

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Fig. 2.15. Extreme air temperature—hot or cold—can become hazardous quickly.

STUFF THAT'S TOXIC

CHEMICAL

Chemical hazards are generally associated with substances that create physical hazards or damage human health. Substances that are corrosive, asphyxiant, or produce harmful vapors, dust, or fumes are all examples. Capacity for chemical hazards may include PPE, spill-containment kits, and chemical-specific neutralizers.

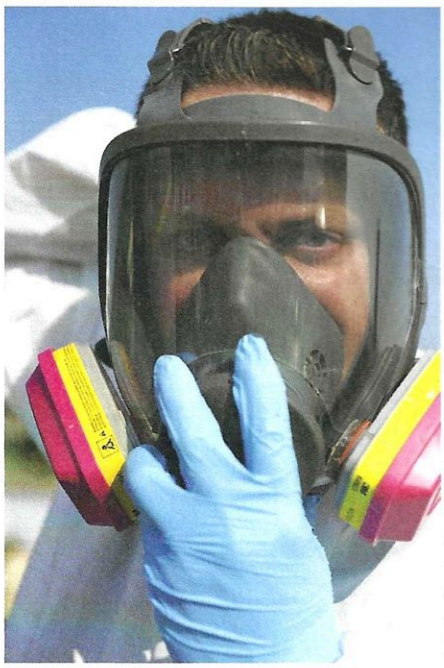


PHOTO BY MICHAELA SLANCHIK FOR UNITED STATES AIR FORCE

Fig. 2.16. Breathing PPE may be required for many situations: dust, paint, fossil fuels, exhaust, confined space, sewage, silica, and more.

Sample capacity questions for chemical:

- Is the safety data sheet (SDS) available and has it been reviewed?
- What hazmat permits are needed for this job?
- What type of PPE suit should be worn?
- What monitoring systems and personnel will be in place?
- How effective are the ventilation procedures?
- What respiratory equipment is appropriate for the work?
- How well equipped are crews for working in confined spaces (such as lights, harnesses, appropriate breathing gear)?
- How accessible are the spill-containment kits?
- Which chemicals have harmful vapors?
- What is the appropriate PPE for working with corrosive chemicals (such as gloves and face shields)? Is that enough?
- Who's checking hydraulic and other chemical hoses for leaks?

BIOLOGICAL

Living organisms present biological hazards in the environment. Animal and insect bites or stings, poisonous plants, bacteria, viruses, bloodborne pathogens, improperly handled food, and contaminated water are all examples. Capacity for these hazards may include bite and sting kits, PPE, poison control kits, and first aid with medication (such as epinephrine pens).

STUFF THAT'S LIVING

Sample capacity questions for biological:

- Where might workers run into hazardous plants (such as poison ivy) and animals (such as rats, spiders, and stinging insects)?
- How can workers protect against animals that could pose a threat (such as dogs, raccoons, and snakes)?
- How do we protect workers against manmade biohazards (such as used syringes and human waste) when reaching into vaults or similar containers?
- How accessible are the first aid resources (such as epinephrine pens and poison control kits)?

Fig. 2.17. Biological hazards come in a variety of forms.

STUFF THAT'S RADIOACTIVE

RADIATION

Radiation hazards exist wherever elements that emit ions or atomic particles exist. Examples include sunlight, radio waves, infrared radiation, and ionizing radiation sources like x-rays or nuclear radiation. While it is uncommon to encounter some of these hazards on most job sites, they can be incredibly damaging to exposed workers. Capacity may include radiation barriers, containment zones, and specialized PPE.

Sample capacity questions for radiation:

- How well equipped are workers to protect against ultraviolet rays (such as sunscreen and sunglasses)?
- Where are the high-powered radio frequency (RF) tower transmitters that workers need to be aware of?
- Where do we have specialized PPE for working near RF transmitters?
- What specialized PPE is required for working around nuclear facilities?

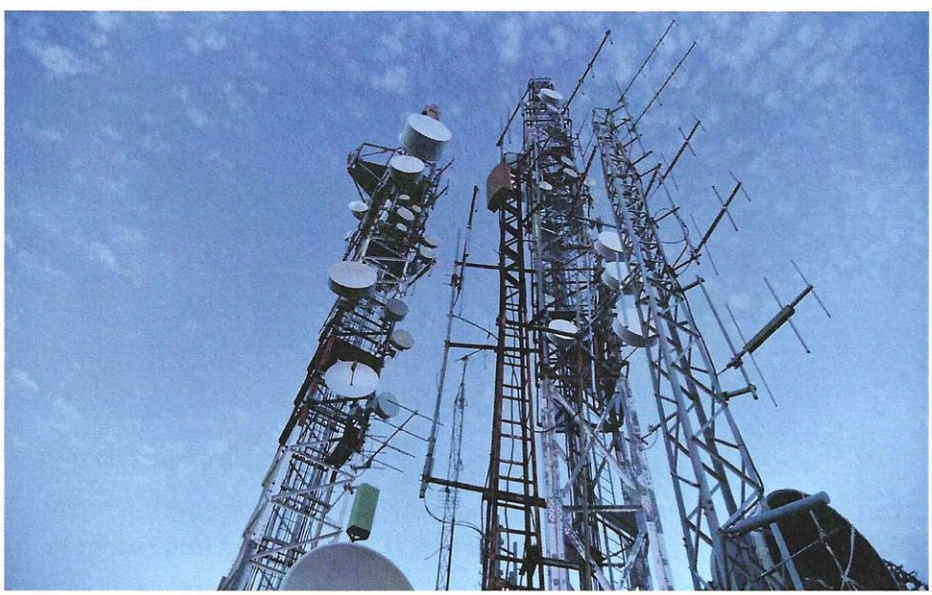
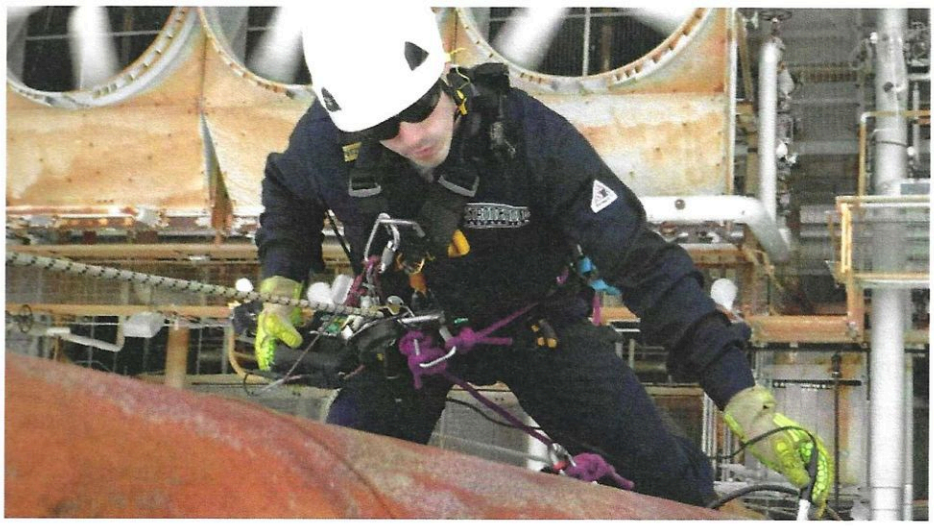


PHOTO BY PEXELS.COM

Fig. 2.18. Radiation hazards come in many forms.



STUFF THAT'S LOUD

Fig. 2.19. Hearing can be damaged by sudden loud noises and constant lower-volume sounds.

SOUND

Sound hazards, while typically not associated with fatalities, can be life-altering hazards nonetheless. Very loud noises in short duration (like explosions) or loud noises over a sustained time can both cause irreversible hearing damage. Sound can also affect workers' ability to communicate with each other when hazardous situations emerge. Capacity for these hazards include PPE and alternative communication methods such as hand signals or other visual indicators.

Sample capacity questions for sound:

- What types of proper hearing protection are onsite?
- How do you verify everyone is using the same hand signals and in the appropriate way?
- Where can we place noisy portable motors (such as air compressors and generators) to keep them away from workers?
- How should workers warn their crew before activating noisy equipment?

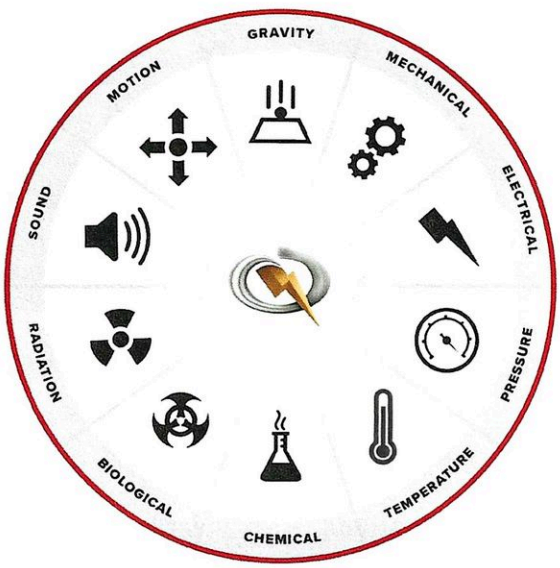
ACTIVITY

JOB BRIEFING

ACTIVITY INSTRUCTIONS

Use the job scenario and photo of the work area below to complete the activity.

1. Discuss the work tasks associated with the job. Use the Task boxes on the facing page to write down three of the tasks associated with the job.
2. For each task, use the Energy Wheel to identify hazards. Write one of these in the Hazard/Energy box.
3. Identify unwanted events that could be a result of that hazard. Write one of the unwanted events in the Unwanted Event box.
4. For each of the identified hazards and unwanted events, discuss preventive controls. List them in the Preventive Controls box.
5. For each of the identified hazards and unwanted events, discuss mitigative controls that are needed to reduce the consequences of the unwanted event. List them in the Mitigative Controls box.



JOB SCENARIO

You are performing live-line work spreading wire for a reconductor job.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

1. Task:

4. Preventive Controls (Prevent the unwanted event):

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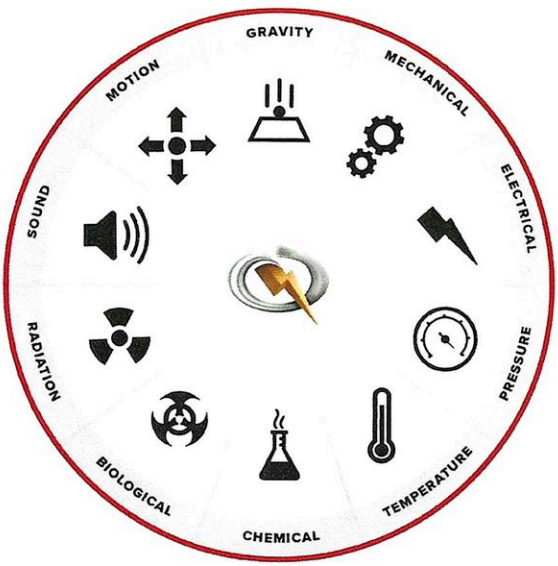

JOB SCENARIO

You are completing a pole changeout project to replace several poles that run alongside a four-lane road and the backyard fences of private homes.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

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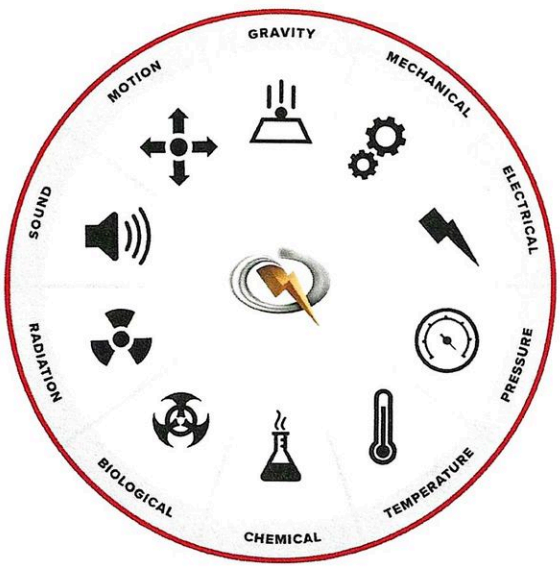

JOB SCENARIO

Your job is to lower a fiber splice using an aerial lift. You will need to block one lane of traffic to lower the splice.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

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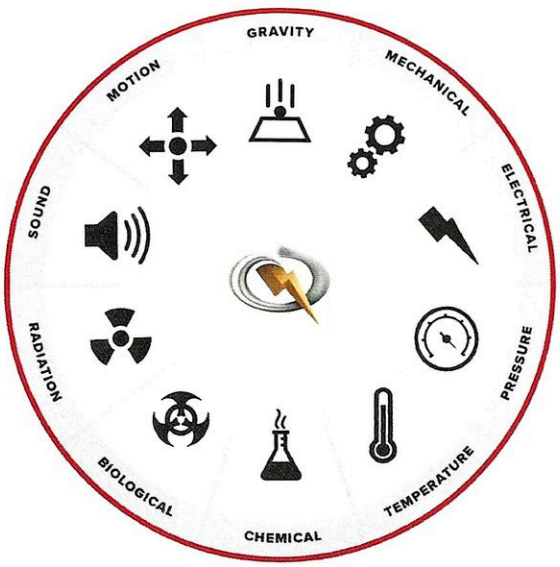

JOB SCENARIO

You are installing new equipment at the top of a cellular tower. You will set up rigging to send the new equipment up.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

NOTES

NOTES

GLOSSARY

capacity	The actual or potential ability to perform, yield, or withstand. With The Capacity Model, the ability to perform work safely, yield safe results, and withstand the outcome of an accident or mistake.
capacity for failure	In a work setting, the ability to perform a task while safely withstanding the outcome of an accident or mistake.
control	Acts (things a person does), objects (a device that works without acts), or systems (a combination of acts and objects) intended to prevent or mitigate an unwanted event.
job briefing	A required meeting held at a minimum before every job or shift to ensure every crewmember understands specific factors and responsibilities for that job. Also referred to as a tailboard.
mitigative control	Act, object, or system designed to reduce the consequences of an unwanted event. Mitigative controls help after the bad stuff has already happened.
preventive control	Act, object, or system designed to prevent an unwanted event. Preventive controls stop the bad stuff from happening.
Quanta Energy Wheel	A hazard-identification tool that helps workers systematically look for types of energy. The Energy Wheel focuses on hazards that can cause a life-threatening, life-altering, or life-ending injury.

STKY	Stuff that kills you. Hazards that can cause a life-altering, life-threatening, or life-ending injury.
STKYa	*A* stands for "actual." A work incident (STKY event) that resulted in an injury that was life-threatening, life-altering, or life-ending.
STKY control	A control that prevents or mitigates a STKY event (life-threatening, life-altering, or life-ending event). Like preventive and mitigative controls, it is an act, object, or system.
STKY discussion	Conversations meant to generate an understanding of how to identify STKY using the Energy Wheel. STKY discussions happen before the start of a job and as conditions change that may impact worker safety.
STKY event	An event where preventive controls failed, resulting in the unexpected or uncontrolled release of energy that could have reasonably resulted in a life-threatening, life-altering, or life-ending event.
STKYp	*P* stands for "potential." A STKY event in which no one was seriously injured but mitigative controls were either not in place, not functioning, or not effective. The only thing protecting the crew from a STKYa was luck.
STKY success	A situation where mitigative controls were in place, functioning, and effective. These controls successfully prevented a STKY event from becoming a STKYa or STKYp.
unwanted event	An event with a potential for harm from exposure to a hazard that was not adequately controlled—it is the bad stuff that can happen because of a hazard.

SOURCES

CONTENT

1. Albert, Alex, Hallowell, Matthew R., Skaggs, Michael, Kleiner, Brian (2017). Empirical measurement and improvement of hazard recognition skill. *Safety Science*. 93: 1-8.

2. Bilotto, Rick (2015). Spotlight on Hazard Recognition. Construction Industry Institute. Retrieved from https://www.construction-institute.org/blog/2015/spotlight-on-hazard-recognition

3. Schenkein, Joyce (2016). Our Brains Are Built To Ignore Things. Here's Why That's Great News. *Forbes*. Retrieved from https://www.forbes.com/sites/quora/2016/01/26/our-brains-are-built-to-ignore-things-heres-why-thats-great-news/#a3689f61ede9

4. Jabr, Ferris (2014). How the Brain Ignores Distractions. *Scientific American*. Retrieved from https://www.scientificamerican.com/article/how-the-brain-ignores-distractions/

5. Bailey, Regina (2019). Overview of the Medulla Oblongata. ThoughtCo. Retrieved from https://www.thoughtco.com/medulla-oblongata-anatomy-373222

6. Siddiqui, Shazia Vegar, Ushri Chatterjee, Devvarta Kumar, Aleem Siddiqui, and Nishant Goyal (2008). Neuropsychology of prefontral cortex. *Indian Journal of Psychiatry*. 50(3): 202–208.

7. Albert, Alex (2013). *Strategies for Construction Hazard Recognition*. Retrieved from https://www.academia.edu/38605687/STRATEGIES_FOR_CONSTRUCTION_HAZARD_RECOGNITION

GRAPHICS

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1.3. Right photo: Malta Dynamics. https://maltadynamics.com/product/tool-lanyard/.

APPENDIX

APPENDIX A—ONLINE LEARNING FOR THE CAPACITY MODEL

The Online Learning for The Capacity Model website provides valuable educational resources. The site is open to everyone. There is no registration requirement. Use it as a resource to refresh and extend your learning.

ACCESS THE WEBSITE

Go to: https://learning.thecapacitymodel.com.

ELEARNING COURSES

Courses are available in English, Spanish, and French. Options include:

- Understanding The Capacity Model
- STKY Controls for Crews
- Introduction to Learning Teams

TCM VIDEOS

The videos provide viewers with an introduction to several elements of The Capacity Model. Some options include Quanta Human Performance Principles, STKY, Quanta Energy Wheel, and Quanta's Safety Philosophy.

STKY RE-CREATION VIDEOS

The re-creation videos are based on actual STKY events. Use these videos to help reinforce concepts such as using the Energy Wheel, asking capacity questions, identifying preventive and mitigative controls, and classifying STKY events.

RESOURCES

The site's Resources section includes PDFs of printable informational brochures and job aids such as the STKY Discussion Guide.

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